

Introductory Logarithms

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1 Introduction

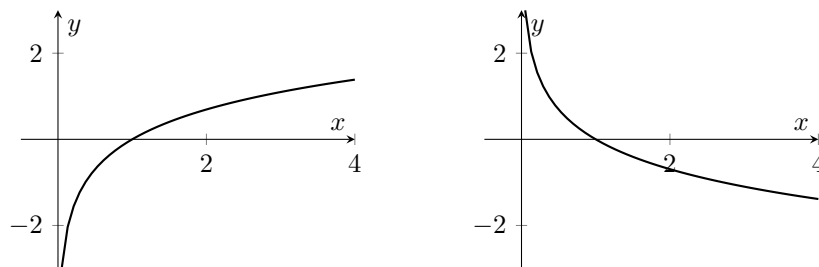
Logarithms answer a simple question: to what power must a fixed **base** b (where $b > 0$, $b \neq 1$) be raised to produce a given value? Formally,

$$b^x = y \iff \log_b(y) = x.$$

This inverse relationship with exponentiation is what makes logarithms useful — operations that are difficult in the multiplicative world, such as multiplying large numbers or computing roots, become simple additions and divisions in the logarithmic world. In this paper, we explore the key properties of logarithms and develop techniques to compute and manipulate them efficiently.

2 Graph of Logarithmic Functions

Consider $y = \log_b(x)$. The shape of the graph depends on whether the base $b > 1$ or $0 < b < 1$.



The left graph shows $y = \log_b(x)$ for $b > 1$: the function is increasing and passes through $(1, 0)$. The right graph shows $y = \log_b(x)$ for $0 < b < 1$: the function is decreasing but still passes through $(1, 0)$. In both cases, the graph approaches the y -axis but never touches it, since $\log_b(x)$ is undefined for $x \leq 0$.

3 Some Properties of Logarithms

We will first go ahead and list the **properties** and then work with their **proofs** in the next section.

The properties below follow directly from the corresponding rules of exponents, since a logarithm is simply an inverse exponent. Throughout, **assume** $b > 0$, $b \neq 1$, **and** $x, y > 0$.

3.1 Product Rule

$$\log_b(xy) = \log_b(x) + \log_b(y)$$

Multiplying two numbers corresponds to adding their logarithms.

3.2 Quotient Rule

$$\log_b\left(\frac{x}{y}\right) = \log_b(x) - \log_b(y)$$

Dividing two numbers corresponds to subtracting their logarithms.

3.3 Power Rule

$$\log_b(x^n) = n \log_b(x)$$

Raising a number to a power corresponds to multiplying its logarithm by that power.

3.4 Change of Base

$$\log_b(x) = \frac{\log_k(x)}{\log_k(b)}$$

for any valid base k . This is especially useful when a calculator or table only supports specific bases, such as 10 or e .

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3.5 Additional Identity: Very Important

$$a^{\log_b c} = c^{\log_b a}$$

Remark. These properties allow logarithms to convert multiplication into addition, division into subtraction, and exponentiation into multiplication — the very reason they were historically used to simplify calculations.

4 Derivations of the Properties

4.1 Product Rule

Let $\log_b(x) = m$ and $\log_b(y) = n$. By definition of the logarithm,

$$x = b^m, \quad y = b^n.$$

Multiplying these,

$$xy = b^m \cdot b^n = b^{m+n}.$$

Taking $\log_b$ of both sides,

$$\log_b(xy) = m + n = \log_b(x) + \log_b(y).$$

4.2 Quotient Rule

Again let $\log_b(x) = m$ and $\log_b(y) = n$, so $x = b^m$ and $y = b^n$. Dividing,

$$\frac{x}{y} = \frac{b^m}{b^n} = b^{m-n}.$$

Taking $\log_b$ of both sides,

$$\log_b\left(\frac{x}{y}\right) = m - n = \log_b(x) - \log_b(y).$$

4.3 Power Rule

Let $\log_b(x) = m$, so $x = b^m$. Raising both sides to the power n ,

$$x^n = (b^m)^n = b^{mn}.$$

Taking $\log_b$ of both sides,

$$\log_b(x^n) = mn = n \log_b(x).$$

4.4 Change of Base

Let $\log_b(x) = m$, so $b^m = x$. Take $\log_k$ of both sides, for any valid base k :

$$\log_k(b^m) = \log_k(x).$$

By the power rule (derived above), the left-hand side becomes $m \log_k(b)$, so

$$m \log_k(b) = \log_k(x) \implies m = \frac{\log_k(x)}{\log_k(b)}.$$

Since $m = \log_b(x)$, this gives

$$\log_b(x) = \frac{\log_k(x)}{\log_k(b)}.$$

4.5 Interchange Property

We derive $a^{\log_b c} = c^{\log_b a}$. Take $\log_b$ of the left-hand side:

$$\log_b (a^{\log_b c}) = (\log_b c)(\log_b a),$$

using the power rule. Take $\log_b$ of the right-hand side:

$$\log_b (c^{\log_b a}) = (\log_b a)(\log_b c),$$

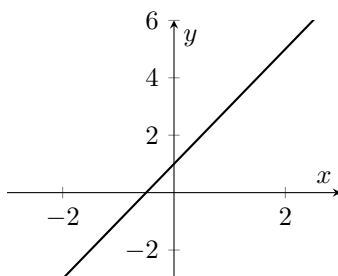
also using the power rule. Since multiplication is commutative, both sides give the same value:

$$(\log_b c)(\log_b a) = (\log_b a)(\log_b c).$$

Because $\log_b$ is a one-to-one function, equal logarithms imply equal arguments, so

$$a^{\log_b c} = c^{\log_b a}.$$

One to One functions: Basically functions where each input has strictly and strictly only one output. Eg. Linear functions of form $y = mx + c$ whenever m is defined and not equal to 0.



THE END
